

Photoelectric effect explained

Notes

The photoelectric effect is the process of emission of electrons from a metal surface when that surface is irradiated with light in the range from infrared to ultraviolet. It is classed as a low energy phenomenon.

Heinrich Hertz (1887) noticed, in his experiments with electrical discharge, that electrodes more readily produced an arc when they were irradiated with UV light.

Observations made and experiments conducted on these effects were in direct conflict with Maxwell's Electromagnetic Wave Theory, which assumed that the production, transmission and absorption of electromagnetic radiation could be described by using the properties of continuous waves.

Planck postulated light is radiated or absorbed in packets (called them quanta), the energy of which is proportional to their frequency: $E = hf$

Albert Einstein used this to explain the photoelectric effect and why it did not obey the wave theory. He published a paper in 1905, for which he received the Nobel Prize in 1921, in which he explained the emission of photoelectrons being caused by light having discrete quantised packets of energy (following on from the Quantum Theory of Planck) and behaving as particles, not waves. His explanation of the photoelectric effect was therefore that of a particle collision.

Experimental evidence shows:

1. For photoelectrons to be emitted from a metal surface there is a minimum frequency (energy) needed in the incident radiation below which no photoelectric emission will occur. This frequency is called the threshold frequency for that metal. It is different for all metals.
e.g. Caesium – yellow – 5.16×10^{14} Hz
Potassium – green – 5.53×10^{14} Hz
Calcium – violet – 6.93×10^{14} Hz
Magnesium – ultraviolet – 8.83×10^{14} Hz
2. The kinetic energy of the emitted photoelectrons is independent of the intensity of the incident light.
3. An increase in intensity of the incident radiation causes more photoelectrons to be emitted. The photoelectric current is directly proportional to the intensity of the incident light provided its frequency is above the threshold frequency.
4. The higher the frequency of the incident radiation above the threshold frequency the greater the number of photoelectrons emitted.

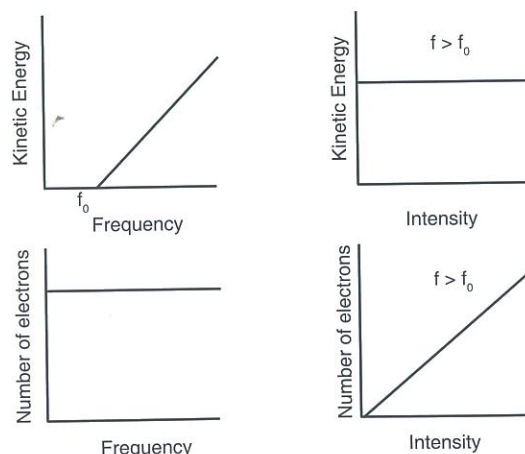


Figure 12.3: Characteristics of the photoelectric effect

The table below gives the work function for a number of surfaces - both in joules and in electron volts. The threshold frequency for each surface is also included.

Element	W (Joules)	W (eV)(V)	f_0 (frequency) (Hz)	λ_0 (wavelength) (nm)
Sodium	3.8×10^{-19}	2.40	5.8×10^{14}	520
Caesium	3.0×10^{-19}	1.88	4.5×10^{14}	666
Lithium	3.7×10^{-19}	1.88	5.6×10^{14}	560
Calcium	4.3×10^{-19}	2.69	6.5×10^{14}	462
Magnesium	4.3×10^{-19}	3.69	8.9×10^{14}	337
Silver	7.6×10^{-19}	4.75	11.14×10^{14}	263
Platinum	10.0×10^{-19}	6.75	15.1×10^{14}	199

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Experiment 12.11: The photoelectric effect

Notes

In 1905, during his “miracle year,” Albert Einstein published five papers. These included special relativity, which dealt with space and time, as well as general relativity, which related mass and energy through the equation $E = mc^2$. However, Einstein won his only Nobel Prize for work he did that same year on the photoelectric effect.

At the time, it was known that light shining on certain materials could knock out electrons to produce a current. It stood to reason that the stronger the light, the greater the current. Researchers also found that how much of a kick the electrons got (or how much kinetic energy they had) depended on the color of the light. Many scientists expected a stronger light would also release an electron with greater energy.

It took Einstein’s brilliance to understand why the color (or frequency) of the light played such a key role in determining how much energy the electrons came away with. The consequences of this insight, along with the contributions of many other scientists, lead to the development of quantum mechanics, which is the basis for the modern electronic world.

This experiment introduces you to the photoelectric effect and guides you to recreate the type of data Einstein interpreted.

Equipment

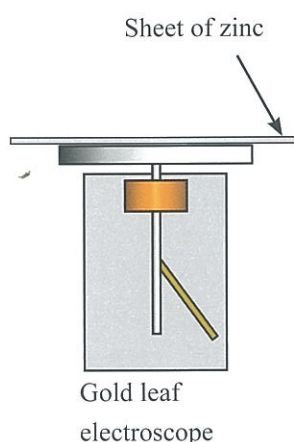
- piece of zinc metal
- short lead/wire
- source of visible light (incandescent lamp)
- source of ultraviolet light - carbon arc lamp or a strong “black light” or laser pointer/pen
- sandpaper or steel wool
- plate of glass
- electroscope (gold leaf)

Lab notes

1. Rub the piece of zinc with a piece of sandpaper or steel wool. This removes oxides to expose the metal.
2. Discharge the electroscope by touching your finger to the electrode.
3. Using a very short lead/wire, attach the zinc to the electroscope.
4. Darken the room.
5. Shine the light from an ultraviolet source onto the zinc.
6. Observe the effect on the electroscope leaves.
7. Discharge the electroscope and shine visible light onto the zinc.
8. Observe the effect of shining the ultraviolet source through a pane of glass that transmits mostly visible range light, but hardly any ultraviolet light.
9. Charge the electroscope positively using a charged glass rod and observe the effect of shining ultraviolet light on the zinc.
10. Charge the electroscope negatively using a charged polythene rod and observe the effect of shining the ultraviolet light on the zinc.

Post lab discussion

1. Compare the effect of the ultraviolet source and the visible source.
2. compare the effect of shining the ultraviolet source through a pane of glass that transmits mostly visible range light, but hardly any ultraviolet light.
3. Ultraviolet light shining on a piece of zinc results in a charge separation. This charge causes the leaves of a negatively charged electroscope to separate further and causes the leaves of a positively charged electroscope to come together. This indicates the charge is negative or, more specifically, consisting of electrons. Visible light does not result in this charge being developed in the zinc.



Problem set 12.2: The photoelectric effect

Notes

1. The photoelectric effect was explained by Einstein, using the idea of Planck, that light consisted of photons with discrete energies dependent on their frequency. Explain how the photoelectric effect is used as evidence for the particle nature of light.
2. An experiment on the photoelectric effect was conducted and the maximum energy of the emitted photoelectrons measured at two different frequencies. When a wavelength of 4.00×10^{-7} m was used the maximum energy of the emitted electrons was 1.40×10^{-19} J and with an incident wavelength of 3.00×10^{-7} m the energy was 3.06×10^{-19} J. From these data derive a value for Planck's constant.
3. Green light with a frequency of 6.7×10^{14} Hz is incident on a caesium metal surface with a work function 2.14 eV. What is the maximum kinetic energy of an electron emitted from this surface as a result of this collision?
4. The work function of aluminium is 4.08 eV.
 - (a) What does the term "work function" mean in this context?
 - (b) What is the maximum wavelength of photons incident on this surface that will cause photoelectrons to be emitted?An aluminium surface is irradiated with ultraviolet radiation of frequency 2.39×10^{15} Hz.
 - (c) Calculate the maximum kinetic energy of the photoelectrons emitted from this surface under these conditions.
5. Whether electrons are, or are not, emitted from a metal surface that is illuminated by light depends on certain factors. State how, if at all, the following properties or factors affect the emission of photoelectrons from a metal surface.
 - (a) The thickness of the piece of metal.
 - (b) The area of surface illuminated.
 - (c) The length of time for which the surface is illuminated.
 - (d) The type of metal the surface is made from.
 - (e) The wavelength of the incident light.
 - (f) The intensity of the incident light beam.
 - (g) The cleanliness of the surface.
6. A nickel surface, work function of 5.01 eV, in an evacuated chamber is irradiated with light of wavelength of 325 nm.
 - (a) What stopping potential will prevent the emitted photoelectrons from travelling between the pair of plates in the chamber?
 - (b) The intensity of the incident light is doubled but the frequency remains constant. What effect will this have on the stopping potential? Explain your answer.

Problem set 12.2: The photoelectric effect

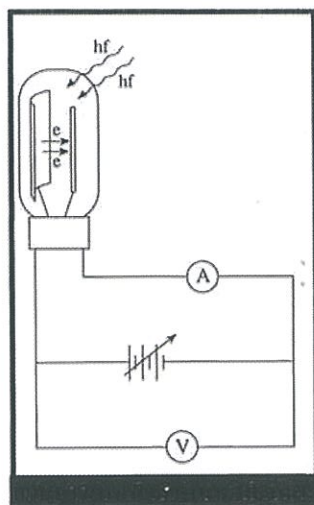
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7. The threshold frequency for aluminium is 1.25×10^{14} Hz.
 - (a) Calculate the work function of aluminium in both joules and electron-volts. An aluminium surface was then irradiated with blue light with a frequency of 6.95×10^{14} Hz and the resultant emitted electrons analysed. Calculate, showing all working:
 - (b) The wavelength of the incident blue light.
 - (c) The energy of the incident photons in both joules and electron-volts.
 - (d) The maximum kinetic energy of the emitted photo-electrons
 - (e) The maximum velocity of the emitted photo-electrons.
8. The surface of a piece of cadmium is irradiated with electromagnetic radiation of wavelength 287 nm. Photoelectric emission stops when a reverse potential of 3.68 V is applied between the anode and cathode of the photoelectric cell.
 - (a) From this data calculate the work function of cadmium in joules.
 - (b) Calculate the maximum velocity of electrons emitted.
9. An experiment is conducted where the surfaces of both platinum and sodium are irradiated with blue light of wavelength 465 nm. Photoelectrons are found to be emitted from the sodium surface but not the platinum surface.
 - (a) Explain why this different effect has occurred with the two metal surfaces. Explain the effect each of the following changes would have on the photoelectric current for each of the metals.
 - (b) The intensity of the incident beam is increased.
 - (c) The blue light was replaced with a light of much longer wavelength.
 - (d) The blue light was replaced with a light of much shorter wavelength.
 - (e) A small reverse potential is applied between the anode and cathode of the photoelectric tube.
10. Ultraviolet light with a wavelength of 3.55×10^{-7} m from a mercury vapour lamp strikes a clean metal surface with a work function of 2.64×10^{-19} J and causes photoelectrons to be emitted.
 - (a) What is the frequency of the incident ultra-violet radiation?
 - (b) What is the energy of a single photon incident on this surface?
 - (c) What is the maximum kinetic energy any photoelectron emitted from this surface can have?
 - (d) What is the longest wavelength of radiation that will eject photoelectrons from this surface?
11. One type of smoke detector contains a photocell, a device that reacts to the amount of light falling on it, and a small permanent light source. If smoke enters the detector, reducing the intensity of light falling on the photocell below a minimum value, the current change in the circuit will set off the alarm. How does the presence of smoke affect the emitted photoelectrons? Does it change their number or their energy or both? Explain.

12. The following data comes from an experiment using a photocell coated with magnesium. The frequency of the incident light was varied and the maximum kinetic energy of the emitted photoelectrons was determined at each frequency.

Frequency	Kinetic energy (J)
1.0×10^{15}	6.0×10^{-18}
1.2×10^{15}	1.9×10^{-19}
1.4×10^{15}	3.3×10^{-19}
1.6×10^{15}	4.6×10^{-19}
1.8×10^{15}	5.3×10^{-19}
2.0×10^{15}	7.2×10^{-19}

- (a) Plot a graph of kinetic energy against frequency.
(b) From the graph determine each of the following.
(i) The threshold frequency of magnesium.
(ii) An experimental value for Planck's constant.
(iii) The work function of magnesium.



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